

SHAHAD ABIR AKASH

Computer Science Undergraduate | Full-Stack & ML Developer

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SUMMARY

Motivated Computer Science undergraduate at CUET with strong problem-solving skills and rapid learning ability. Experienced in building full-stack web applications and real-time systems using React, Node.js, and Socket.io. Proficient in machine learning frameworks like PyTorch and scikit-learn. Seeking internship or junior software engineer roles to apply full-stack and ML skills.

KEY SKILLS

- Languages: C, C++, JavaScript, Python, Bash
- Frontend: React, React Native, HTML, CSS
- Backend & Real-time: Node.js, Express.js, Socket.io, REST APIs
- Databases: MySQL, MongoDB (NoSQL), Appwrite
- DevOps & Microservices: Docker, Nginx, CI/CD Pipelines, Cloud Deployment, Microservices Architecture
- Monitoring & Observability: Prometheus, Grafana, Elastic Stack (ELK), Kibana, Jaeger
- ML / Data: NumPy, pandas, scikit-learn, PyTorch
- Tools: Git, GitHub, Linux
- Other: Problem solving, debugging, agile development

RESEARCH INTERESTS

- Generative models and diffusion-based image synthesis
- Language Models (LLMs)
- learning systems and optimization for large models
- time distributed systems and WebSocket architectures
- backend and cloud-native applications
- DevOps architecture
- Data structures and algorithms

PROJECTS

Diffusion Model Fine-tuning (DDPM)
PyTorch, Hugging Face Diffusers, DDPM/DDIM, torchvision
 2025

Fine-tuned pretrained DDPM UNet on custom butterflies dataset. Implemented DDIM sampling, gradient accumulation, and torchvision preprocessing for improved efficiency and quality.

DCGAN Face Generation (CelebA)
PyTorch, GANs, Deep Convolutional Networks
 2025

Built DCGAN to generate photorealistic faces from 200k+ CelebA images. Addressed mode collapse and training stability through architectural adjustments and regularization.

Video Action Recognition (UCF101)
PyTorch, R(2+1)D CNN, OpenCV, 3D Convolutions
 2024

Implemented R(2+1)D model with transfer learning from Kinetics-400 on UCF101 dataset, achieving strong performance in temporal action classification.

Heart Disease Prediction System
Python, scikit-learn, pandas, NumPy, Matplotlib

📅 2024

Built and compared multiple classifiers on clinical data. Achieved 85–90% accuracy, providing reliable predictive insights.

Full-Stack E-Commerce Website

React, Node.js, MySQL, Express.js, SSLCommerz

📅 2024

Developed complete platform with product listing, cart, and checkout. Integrated SSLCommerz for secure payments and designed REST APIs.

Social Media Web App

React, Appwrite, Appwrite Database, TanStack Query

📅 2023

Built platform with user authentication, post creation, and infinite scrollable feeds. Leveraged Appwrite BaaS and TanStack Query for efficient caching and real-time updates.

Real-Time Messaging App

React, Node.js, Socket.io, MongoDB

📅 2023

Developed robust chat app supporting one-on-one and group messaging. Implemented WebSocket communication via Socket.io for low-latency interactions.

RELEVANT COURSEWORK

Core Computer Science

Data Structures, Algorithms, Object-Oriented Programming, Database Systems, Operating Systems

Systems & Networks

Computer Networks, Distributed Systems, Microprocessor Systems, Linux Programming

Artificial Intelligence & Machine Learning

Artificial Intelligence, Machine Learning, Deep Learning, Pattern Recognition

Mathematics for Computing

Linear Algebra, Probability & Statistics, Discrete Mathematics, Numerical Methods

ACHIEVEMENTS

- Solved 300+ problems on LeetCode and other online platforms.
- Participated in regional Bangladesh Physics & Math Olympiad.
- Microservices Architecture Hackathon participant.
- Multiple hackathon finalist.
- Member of Lions Club Chittagong.

EDUCATION

B.Sc. in Computer Science

Chittagong University of Engineering & Technology (CUET)

📅 Expected 2027

📍 Chittagong, Bangladesh

HSC, Science

Gov. Hazi Muhammad Mohsin College

📅 2021

📍 GPA: 5.0

SSC, Science

Khaja Ajmery High School

LANGUAGES

- Bengali (Native)
- English (Intermediate B1)

VOLUNTEER EXPERIENCE

Member, Lions Club Chittagong

Ongoing

Participated in community service programs focused on social welfare and public outreach. Assisted in organizing charity events and volunteer initiatives. Collaborated with team members to plan, coordinate, and execute activities.

OPEN SOURCE CONTRIBUTIONS

Maintained multiple public GitHub repositories focused on backend systems, real-time communication, and ML pipelines. Developed and published reusable components for Node.js services and React-based applications. Contributed bug fixes, documentation improvements, and feature enhancements to open-source projects used in academic and personal work.

REFEREES

Mir Md. Saki Kowsar
Assistant Professor, Dept. of CSE, CUET

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💡 Course Instructor / Project Guide

[Second Referee Name]

[Designation, CUET or Other]

📧 [Email] | [Phone]

💡 [Relationship, e.g., Project Supervisor]